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Saveetha Institute of Medical And Technical Sciences

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COURSE NAME: COMPUTER NETWORK

Question 1: Transport Layer Revolution & Connection Management

1.1 Limitations of TCP :

Traditional HTTP/HTTPS communication has historically used TCP as its transport protocol. A TCP connection is identified by a 4-tuple:

Source IP + Source Port + Destination IP + Destination Port

For example:

Phone IP:5000 → Web Server IP:443

When a smartphone moves from Wi-Fi to 4G/5G, its network address can change.

Before switching:

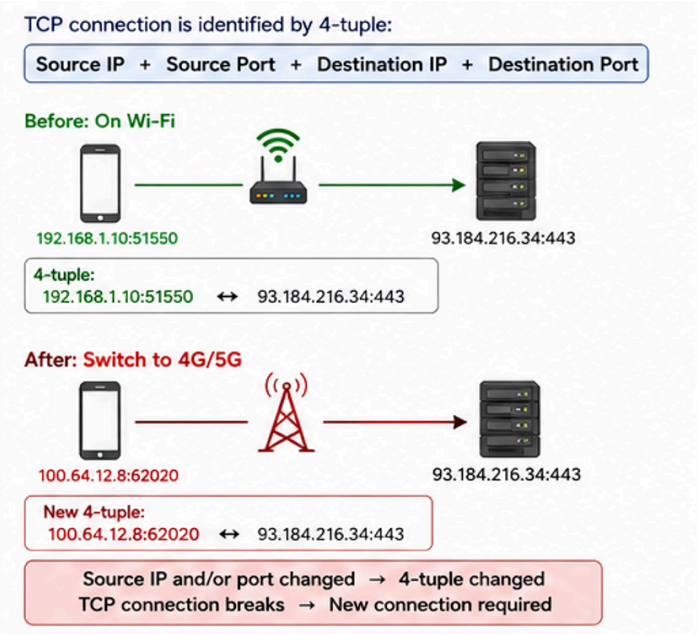
Wi-Fi IP:5000 → Server IP:443

After switching:

4G/5G IP:6200 → Server IP:443

The source IP address and possibly source port have changed, so the new packets no longer match the original TCP 4-tuple.

As a result, the existing TCP connection is broken and the browser may need to establish a new TCP connection to the HTTPS server. This causes additional connection setup delay and can interrupt ongoing web communication.



1.2 QUIC Connection ID Mechanism

The major difference is that QUIC does not depend on the traditional TCP 4-tuple to identify a connection. Instead, QUIC uses a **Connection ID (CID)**.

For example, when a smartphone connects to an HTTPS website using HTTP/3, the QUIC connection may have:

Connection ID = ABC123

Initially:

Wi-Fi IP:5000 → Server:443

The QUIC packets contain:

CID = ABC123

When the user switches to 5G:

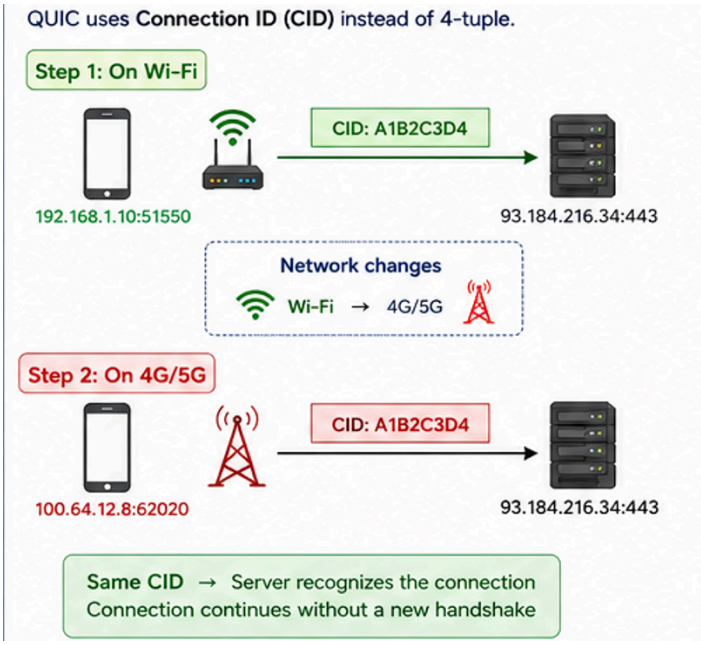
5G IP:6200 → Server:443

Although the **IP address and port have changed**, the QUIC packets can continue using:

CID = ABC123

The server reads the Connection ID and understands that the packets belong to the same QUIC connection.

This mechanism is called **Connection Migration**.



1.3 Why QUIC Uses UDP

QUIC could have been designed as a completely new transport-layer protocol, but this would create a major deployment problem.

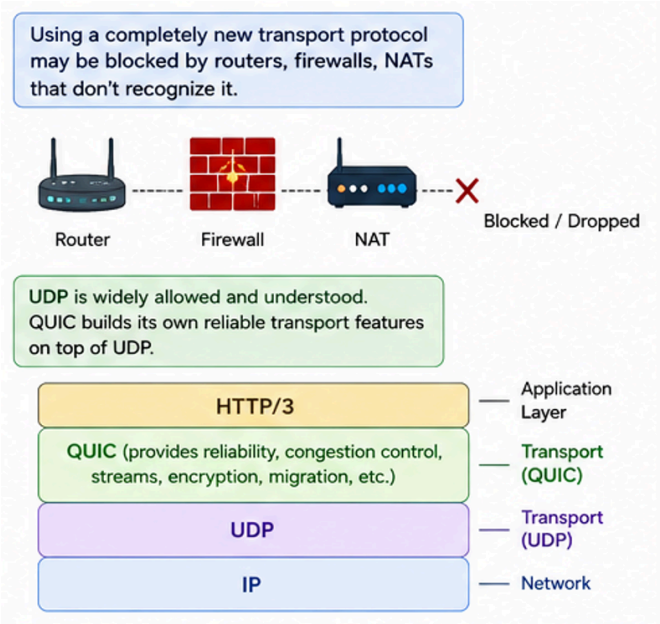
If QUIC had used a completely new IP transport protocol, many existing firewalls and routers might not recognize it and could **block or discard the packets**.

Therefore, QUIC was designed to operate **on top of UDP**.

UDP provides the basic packet delivery mechanism, while QUIC implements its own advanced transport features:

- Reliable data delivery
- Congestion control

- Connection IDs
- Connection migration
- Stream multiplexing
- TLS 1.3 encryption
- Faster connection establishment



Question 2: Head-of-Line Blocking & Streams:

2.1 TCP Head-of-Line Blocking :

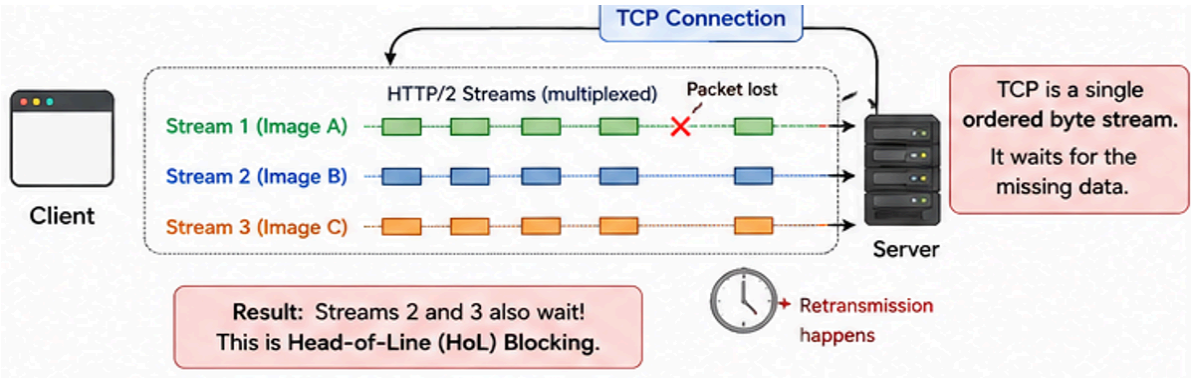
HTTP/2 uses multiplexing, meaning multiple web resources such as HTML, CSS, JavaScript, and images can be sent through a **single TCP connection**.

Suppose a TCP packet containing data for **Image A** is lost.

TCP provides an **ordered byte stream**, so it cannot deliver later data to the application until the missing TCP data has been recovered.

Therefore, even if Image B and Image C packets have already arrived, their data may have to wait for the missing Image A data.

This is called **Head-of-Line (HoL) Blocking**.



2.2 How QUIC Solves Head-of-Line Blocking :

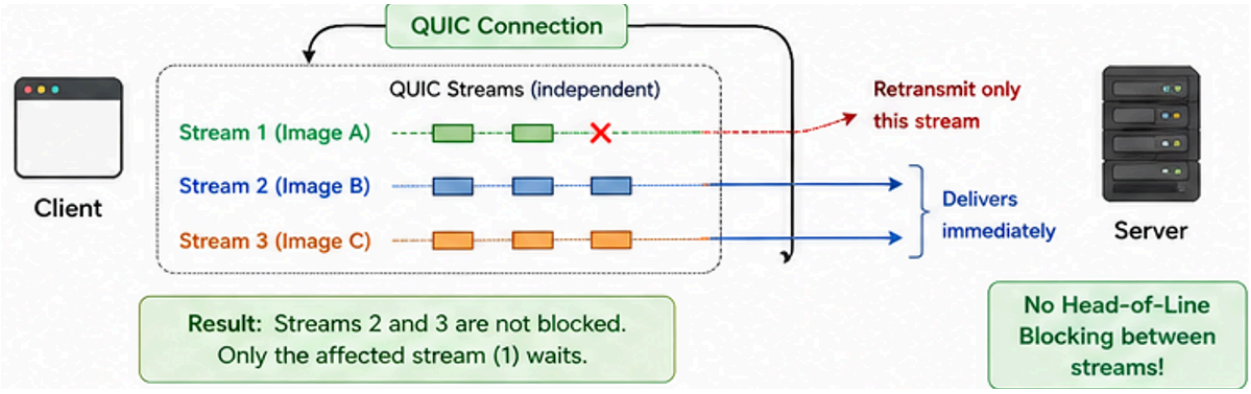
HTTP/3 uses QUIC instead of TCP.

QUIC supports multiple **independent streams** within one connection.

Suppose a packet belonging to **Stream 1 (Image A)** is lost.

QUIC retransmits the missing data for Stream 1, but **Stream 2 and Stream 3 do not have to wait for Stream 1**.

Therefore, the loss of a packet in one QUIC stream does **not block unrelated streams** on the same QUIC connection.



Feature	HTTP/2	HTTP/3
Transport	TCP	QUIC
Lower protocol	TCP	UDP
Encryption	TLS	TLS 1.3 integrated with QUIC
Multiplexing	Yes	Yes
Independent streams	Limited by TCP ordering	Yes
Packet loss effect	Can block other streams	Mainly affects the lost stream
Connection migration	No	Yes, using Connection IDs

